

Sustainable Fishkeeping

Becca Bond, Douglas Sanchez, Sami Wang

Spring 2026

Project description

In this project, we learned how to be knowledgeable and responsible fish owners while minimizing environmental impact. We acquired and practiced essential fishkeeping skills by maintaining an aquarium for a male betta, as well as other aquarium dwellers such as live plants. Through research, troubleshooting and problem-solving, we learned how to maintain a healthy and sustainable aquarium.

Learning Objectives & Achievements

Our first learning objective was to gain confidence and knowledge in fish ownership, including a thorough understanding of:

- Saltwater vs freshwater aquatic environments
- Ideal conditions for an aquarium, including temperature, chemicals, pH level, and light intensity.
- Mechanical aquarium systems for maintaining safe and healthy water, including filtration, heating and aeration.
- Common stresses and diseases contracted by fish and how to treat them.

Our first step towards this objective was choosing the appropriate fish to keep for our project. We wanted a freshwater, hardy fish with light waste, low community needs and a relatively small tank requirement. To do this, we researched nine common species of pet fish and categorized them using a table of traits (table 1). From this, we found that a betta fish would best suit our project goals.

Fish	Tropical/Cold	Fresh/Salt	Sensitivity	Waste	School/Single	Size	Tank min	Price
Betta	Tropical	Fresh	Hardy	Light	Single	2-3"	5 gal	\$4-30
Dwarf gourami	Tropical	Fresh	Hardy	Light	Single	2-3"	10 gal	\$4-20
Goldfish	Cold	Fresh	Hardy	Heavy	School, single	6-10"	55+	\$6-35
Badis	Tropical/Sub	Fresh	Hardy-ish	Light	Single	2-2.5"	5 gal	\$5-20
Platy	Tropical/Sub	Fresh	Hardy	Heavy	Community	2-3"	10 gal	\$2-10
Rasbora	Tropical	Fresh	Hardy	Light	School	1-3"	5-10 gal	\$2-15
Danio	Cold/Temperate	Fresh	Hardy	Heavy	School	1-3"	10 gal	\$2-10
Endler's livebearer	Tropical/Temp	Fresh	Hardy	Light	School	1-2"	10 gal	\$2-10
Tetra	Tropical/Temp	Fresh	Hardy	Light	School	1-1.5"	10-20 gal	\$2-15

(Table 1: Categorization of nine species of fish and their major traits as pets)

After specifying bettas as our project fish, we researched additional traits specific to bettas that would help us maintain proper aquarium conditions. These traits are summarized in table 2.

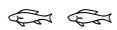
Condition	Value
Temperature	76-82 deg F
pH level	6.8-7.5
Nitrate level	<20 ppm
GH level	70-140 ppm
Environment	Hiding, playing
Food	High protein
Tank size	5+ gallons
Tank mates	Snails, shrimp
Current	Low filter
Aeration	Surface access

(Table 2: Ideal environmental conditions and preferences for a betta fish)

Noticeable aquarium conditions that we learned about during our research were:

- GH, or general hardness, which measures the water's calcium and magnesium concentrations;
- Nitrates, which aren't harmful at low levels but are converted from toxic ammonia and nitrites during the nitrogen cycle;
- Current caused by artificial filtration, which should be kept low for the fish's comfort while still allowing for effective water filtration; and
- The three stages of artificial filtration: mechanical, in which water flows through a sponge to remove large debris; biological, in which water flows through porous, bacteria-housing media to promote the nitrogen cycle; and chemical, in which water flows through activated carbon to remove dissolved impurities, tannins and odors. Each of these steps is important to the effective filtration of aquarium water.

Lastly, we drove to the Waltham Petco and selected a male Veiltail betta who was friendly and full of energy. After giving him a brief tour of Olin's campus and introducing him to his new aquarium home, we named him Sonic.



Our second learning objective was to gain an appreciation for responsible fishkeeping and to develop sustainable practices for our aquarium. To this end, we aimed to:

- Increase energy efficiency through natural filtration, mainly live plants. Plants provide natural filtration of toxins like ammonia, nitrites and nitrates, playing a vital

role in an aquarium's nitrogen cycle. They also naturally control algae and produce oxygen, further improving the aquatic environment.

We utilized volcanic stratum in place of gravel or pebbles to securely anchor plants in the tank and offer them beneficial nutrients. Volcanic substrate is nutrient-rich and lightweight, which helps promote root development and plant growth. In addition, it's environmentally friendly because it's made of natural, volcanic soil.

- Practice responsible waste management by reusing fish waste as plant fertilizer. Fortunately, we didn't experience a great amount of waste from Sonic other than occasional leftover food. We assume the aquarium's mechanical and natural filtration systems took care of the rest.
- Prioritize water conservation and reuse. We performed water changes to regulate ammonia and nitrite levels in the water, especially during the first week as we combatted high nitrite levels caused by the new stratum. The nitrite-filled water, however unhealthy for fish, was very beneficial for plants, so we used the removed water to feed our dorm cacti.



Our final learning objective was to gain real-world skills in experimentation, problem-solving and troubleshooting. As student fishkeepers, our journey with Sonic was filled with learning opportunities. For example, we were unprepared to immediately introduce Sonic to the new aquarium because when we added the stratum, the water turned black with dust and ammonia levels spiked. Sonic stayed in his Petco cup for several days while we learned how to gently replace 25% of the new water each day and add Prime conditioner to stabilize ammonia and nitrite levels. From this, we learned that new experiences may be seen as learning opportunities but should be approached with care when a fish's life is on the line.

A second learning experience we've had with Sonic was training him using positive reinforcement. We trained him to come to our hand by only placing food near our hand during meals; this encouraged him to be comfortable around our hand until he immediately swam to our hand upon seeing it, hoping for a piece of food. We've also begun to move him from his tank to his Petco cup and breather bags (which allow oxygen in and carbon dioxide out even when sealed) and give him food in those. We do this to increase his comfort traveling – soon, Sonic will take summer leave and will travel across the United States to a home away from home. Bon voyage, Sonic!

Progress Photos



Image 1: Sonic's first day on campus; we kept him in his Petco cup while we waited for the new tank's ammonia and nitrite levels to subside. In addition to not having prepared new water, we also fed him too much food while he was in the cup, especially considering his small space and high stress levels. It was a learning experience!



Image 2: Sonic's first day in his new tank. The water was still visibly gray from the stratum dust, but nitrite levels were safe thanks to frequent water replacements and water conditioning using Prime Stability (shown on the right).



Image 3: Putting Sonic back in his Petco cup to introduce him to traveling conditions and prepare him for summer break. We gave him food as positive reinforcement to teach him that the cup was a good place to be.



Image 5: Giving Sonic toys to play with; he was not amused by the green ball and treated the duck as an enemy, flaring his operculum and waving his fins aggressively.

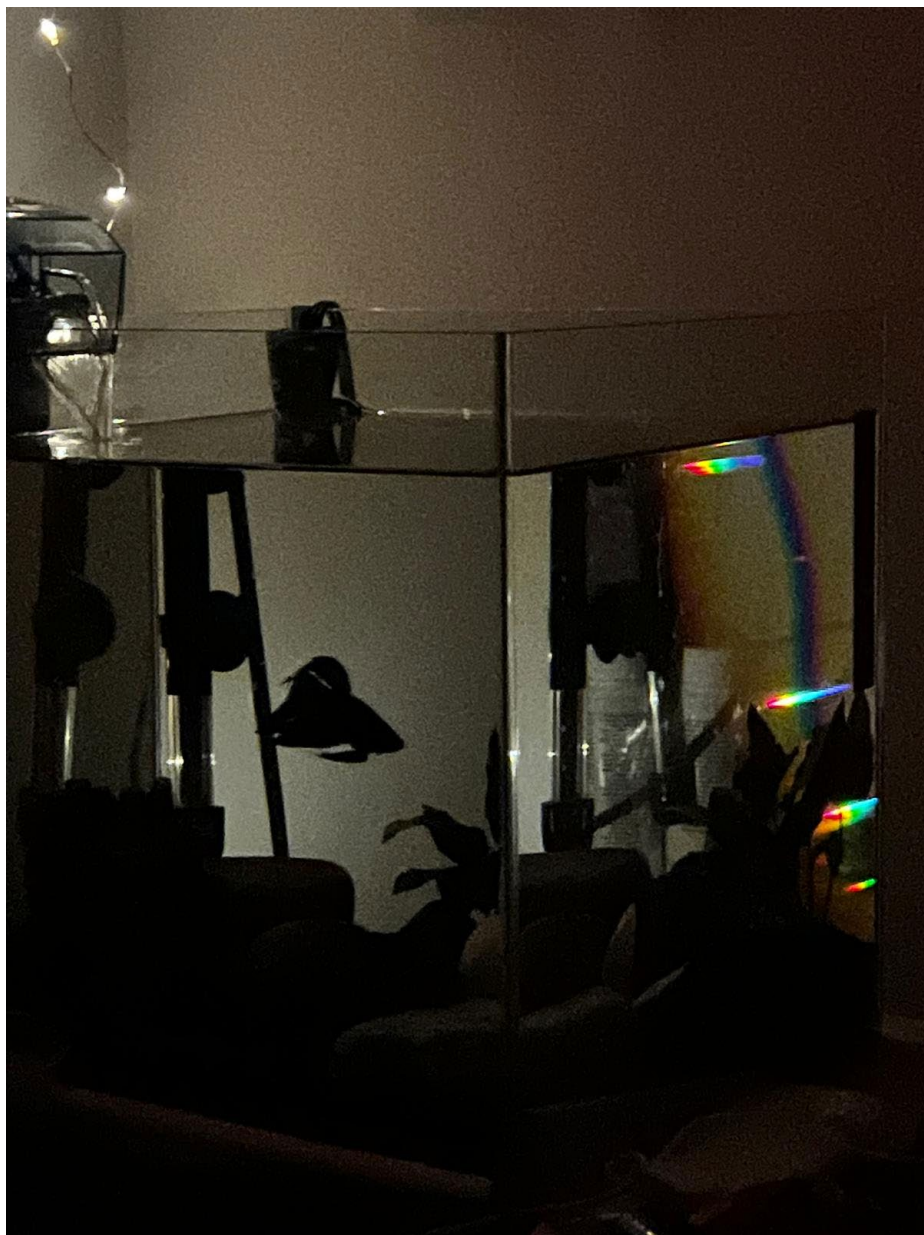


Image 4: Rainbow fish

